

NIALL MCCARTHY

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PROFESSIONAL SUMMARY

Experienced Research Engineer with 14 years of hands-on expertise in machine learning infrastructure, distributed systems, and data engineering. Proven track record of designing and implementing scalable research platforms that support advanced analytics and AI workloads. Seeking challenging opportunities to lead technical innovation in research-driven environments while mentoring junior engineers and driving architectural excellence.

WORK EXPERIENCE

Research Engineer, Staff

Zenith Data Systems (Previously known as CloudBridge Analytics)

Dublin, Ireland

www.zenithdata.io

January 2021 - Present

Architected and deployed machine learning inference platform supporting 50+ concurrent research projects with sub-100ms latency requirements

Led migration of legacy research pipelines from on-premise infrastructure to cloud-native architecture, reducing operational costs by 35%

Established monitoring and observability standards across research infrastructure using Prometheus and Grafana dashboards

Mentored team of 6 junior engineers on distributed systems design patterns and best practices for reproducible research workflows

Senior Research Engineer

Analytics Innovations Ltd

Dublin, Ireland

www.analyticsinnovations.ie

March 2018 - December 2020

Designed and implemented real-time data processing framework handling 2TB of daily research datasets using Apache Spark

Developed comprehensive testing suite for research algorithms achieving 94% code coverage and reducing production incidents by 60%

Collaborated with data scientists to optimize feature extraction pipelines resulting in 40% improvement in model training time

Established CI/CD processes for research code deployments with automated validation and rollback mechanisms

Research Engineer

Titan Research Group

Belfast, Northern Ireland

www.titanresearch.net

June 2010 - February 2018

Implemented statistical analysis tools and visualization frameworks for computational research projects

Developed Python-based automation scripts to streamline data collection and preprocessing workflows for 15+ concurrent studies

Contributed to open research initiatives by documenting technical methodologies and publishing infrastructure design decisions

Managed SQL and NoSQL databases supporting research data warehouses with millions of records and complex query patterns

EDUCATION

Bachelor of Science in Computer Science

University College Dublin

Dublin, Ireland

Graduated: 2010

SKILLS

Programming Languages: Python, Java, Scala, Go, C++

Data Technologies: Apache Spark, Kafka, Hadoop, Hive, SQL, PostgreSQL

ML Frameworks: TensorFlow, PyTorch, Scikit-learn, XGBoost

Infrastructure: Kubernetes, Docker, Terraform, ArgoCD, AWS, Azure

Databases: PostgreSQL, MongoDB, Cassandra, Redis

Tools: Git, Jenkins, Airflow, Jupyter, Pandas, NumPy

Cloud Platforms: AWS, Azure, GCP

Soft Skills: Technical Leadership, Mentoring, Architecture Design, Cross-functional Collaboration